THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For which value of k does $x^2 + kx + 36 = 0$ have equal roots (take $k > 0$)? 1

(a) 36 **(b)** 12 **(c)** 6 **(d)** 24

2. If $x = 3$ is a root of $kx^2 - 5x - 3 = 0$, then $k =$ 1

(a) $\frac{4}{3}$ **(b)** 2 **(c)** $\frac{1}{3}$ **(d)** -2

3. The equation $kx^2 + 10x + 5 = 0$ has real roots when: 1

(a) $k < 5$ only **(b)** $k > 0$

(c) $k \leq 5$ and $k \neq 0$ **(d)** $k \geq 5$

4. If α, β are roots of $x^2 - 8x + 15 = 0$, then $\frac{1}{\alpha} + \frac{1}{\beta} =$ 1

(a) 120 **(b)** $\frac{8}{15}$ **(c)** $\frac{15}{8}$ **(d)** $-\frac{8}{15}$

5. If the roots of $x^2 - 8x + 12 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 52 **(b)** 64 **(c)** 40 **(d)** 88

6. If the roots of $x^2 - 18x + k = 0$ differ by 8, then $k =$ 1

(a) 97 **(b)** 130 **(c)** 65 **(d)** 73

7. For which value of k does $x^2 + kx + 49 = 0$ have equal roots (take $k > 0$)? 1

(a) 14 **(b)** 7 **(c)** 49 **(d)** 28

8. If $x = 2$ is a root of $kx^2 + x - 10 = 0$, then $k =$ 1

(a) 2 **(b)** -3 **(c)** -2 **(d)** $\frac{5}{2}$

9. **Assertion (A):** $(x + 2)^3 = x^3 - 4$ is a quadratic equation. 1

Reason (R): An equation of the form $ax^2 + bx + c = 0$ with $a \neq 0$ is a quadratic equation.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The roots of $2x^2 - 7x + 3 = 0$ are 3 and $\frac{1}{2}$. 1

Reason (R): The quadratic formula gives the roots as $\frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The sum of a number and its reciprocal is $\frac{17}{4}$. Find the number. 2

12. Find the values of k for which $kx^2 + 10x + 5 = 0$ has real and equal roots. 2

13. If α, β are the roots of $x^2 - 10x + 9 = 0$, find $\alpha^2 + \beta^2$. 2

14. The sum of a number and its reciprocal is $\frac{26}{5}$. Find the number. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve for x: $\frac{x+2}{x-2} + \frac{x-2}{x+2} = \frac{58}{21}$. 3

16. For what values of k does $x^2 + 13x + k = 0$ have real roots? Find the greatest integer value of k. 3

17. The sum of the squares of two consecutive odd positive integers is 394. Find them. 3

OR

Solve for x: $\frac{x+1}{x-1} + \frac{x-1}{x+1} = \frac{34}{15}$. 3

18. For what values of k does $x^2 + 15x + k = 0$ have real roots? Find the greatest integer value of k. 3

OR

The sum of the squares of two consecutive odd positive integers is 514. Find them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ananya checks the nature of roots of three equations without solving them: $2x^2 - 6x + 3 = 0$, $3x^2 - 4\sqrt{3}\,x + 4 = 0$ and $x^2 + 2x + 5 = 0$. 5

(i) Find the discriminant of the first equation.

(ii) What is the nature of the roots of the second equation?

(iii) Does the third equation have real roots? Give a reason.

(iv) Find the values of k for which $2x^2 + kx + 3 = 0$ has equal roots.

20. A motorboat goes 40 km upstream and 40 km downstream in 6 hours. If the speed of the stream is 5 km/h, find the speed of the boat in still water. 5

OR

A train travels 720 km at a uniform speed. If the speed had been 4 km/h more, it would have taken 2 hours less for the journey. Find the original speed. 5

END OF QUESTION PAPER — **Total: 40 Marks**